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GPEC 443: GIS and Spatial Data Analysis

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1.1 Data Assembly and Analysis

There are 32 Köppen–Geiger climate zones represented in the *Beck et al. (2018)* data file labeled as *GRIDCODE*. Each climate zone has a corresponding alphanumeric code (e.g., *Af*, *BSh*, *Aw*) that denotes its main climate type, precipitation pattern, and temperature characteristics, as represented on *Figure 1*. To prepare the dataset for analysis in QGIS, I cleaned the data in excel by removing extraneous columns not necessary for spatial comparison or aggregation.

1.2 Conflict Data and Spatial Overlay

Another dataset used in this analysis was the *UCDP/PRIO v25 (2024)* violent conflict event dataset, which includes the geographic coordinates and event counts of violent conflicts. By overlaying the Köppen–Geiger climate data and the UCDP/PRIO event data in QGIS, I was able to spatially identify which climate zones experience the greatest frequency of violent conflict. Using the *Extract by Location* and *Clip* tools, I clipped conflict points to their respective climate zones exclusively in the Northern South America region.

1.3 Spatial Counting and Quantitative Summary

Then, using the *Count Points in Polygon* tool in QGIS, I calculated the total number of violent conflicts that occurred within each distinct Köppen–Geiger climate region. The resulting summary was exported into Excel organized into *Table 1*, which reports the total number of conflicts per climate region represented on the Map (*Figure 1*). I imported the output table into Excel and used a pivot table to further refine the data and calculate cumulative counts by zone. This process allowed for more precise quantitative insight on the visual climate–conflict relationships across Northern South America.

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2. Results and Discussion

While the dataset is global, I focused my spatial analysis on the Northern South America region, near the Panama Canal, because it exhibits a very high density of conflict events in the UCDP/PRIO data. This conflict-dense region was interesting for examining the climate-conflict relationship because there was stark contrasts between climate zones and the presence of violent conflict events.

The results reveal distinct spatial patterns between climate zones and violent conflict frequency across Northern South America. The **equatorial savanna (Aw), equatorial fully humid (Af),** and **equatorial monsoon (Am)** zones exhibit the highest concentration of conflict events, especially in regions with moderate to high rainfall, and notably, agricultural potential. Also, there is strong conflict density close to the neighboring **warm temp fully humid warm summer (Cfb)** zone. *Koubi (2018),* argues that regions experiencing greater precipitation and fertile agricultural land may be at higher risk of conflict due to economic dependence on agriculture and exposure to climate variability.

In contrast, the **arid (BWh and BSh)** zones show much fewer conflict events. This may be attributed to lower population densities and limited agricultural productivity due to arid climate. According to *Koubi (2018)*, this might reduce direct competition for resources in those

regions but also show vulnerability and potential for migration pressures under prolonged drought conditions in these arid regions. Resulting in greater reliance and pressure on the agriculture-friendly zones which are already conflict dominated.

These findings suggest that while arid zones experience less direct conflict, the climates where agricultural livelihoods thrive, face greater sensitivity to instability and risk when climate related shocks occur. As *Chassang and Padró i Miquel (2009)* note, "*adverse climatic conditions can lead to higher food prices by dramatically reducing crop yields and the subsequent supply of crops*." Short-term increases in food prices may, in turn, "*amplify the opportunity cost of rebellion, since they further reduce the short-term opportunity cost of fighting*.". Basically, when extreme weather events or shifts in precipitation reduce crop yields, food insecurity can intensify, political grievances can rise, and social conflict/unrest can ensue.

When such shocks are compounded by weak governance or corruption, public frustration increases as governments fail to meet citizens' needs in times of crisis. The resulting dynamic reinforces the broader climate–conflict hypothesis: that environmental stressors act not as isolated causes of violence but as *threat multipliers* that exacerbate existing socioeconomic and political vulnerabilities.

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3. Conclusion

This analysis demonstrates how spatial data visualization in QGIS can clarify the geographic intersections of climate and conflict. By integrating Köppen–Geiger climate data with conflict event data, I was able to identify climate regions where environmental and social stress tend to converge. The findings support prior research suggesting that climatic variability interacts with political and economic factors to shape patterns of instability. Future research could refine this analysis by incorporating temporal data to further evaluate the timing of these conflicts and understand more specifically if a notable climate event, or political event occurred in the time leading up to these violent conflicts. This would be especially interesting if the dense clusters of conflict occurred around the same time.

References

Beck, H.E., et al. (2018). *Present and future Köppen–Geiger climate classification maps at 1-km resolution*.

UCDP/PRIO (2024). *Uppsala Conflict Data Program / Peace Research Institute Oslo Armed Conflict Dataset, Version 25*.

Koubi, V. (2018). *Climate change and conflict*.

Chassang, S., & Padró i Miquel, G. (2009). *Economic shocks and civil conflict: An instrumental variables approach*.

Table 1

Total number of violent conflict events per Köppen–Geiger climate zone.

Climate Zone	Total Violent Conflicts
11	4872
12	3372
13	333
14	3909
22	3
27	174
32	2491
35	58
38	39
(blank)	
Grand Total	**15251**

Note: *Counts are derived from the UCDP/PRIO v25 (2024) conflict dataset spatially overlaid with Beck et al. (2018) climate zones. The table summarizes conflict frequencies across all zones included in the study region (northern South America).*
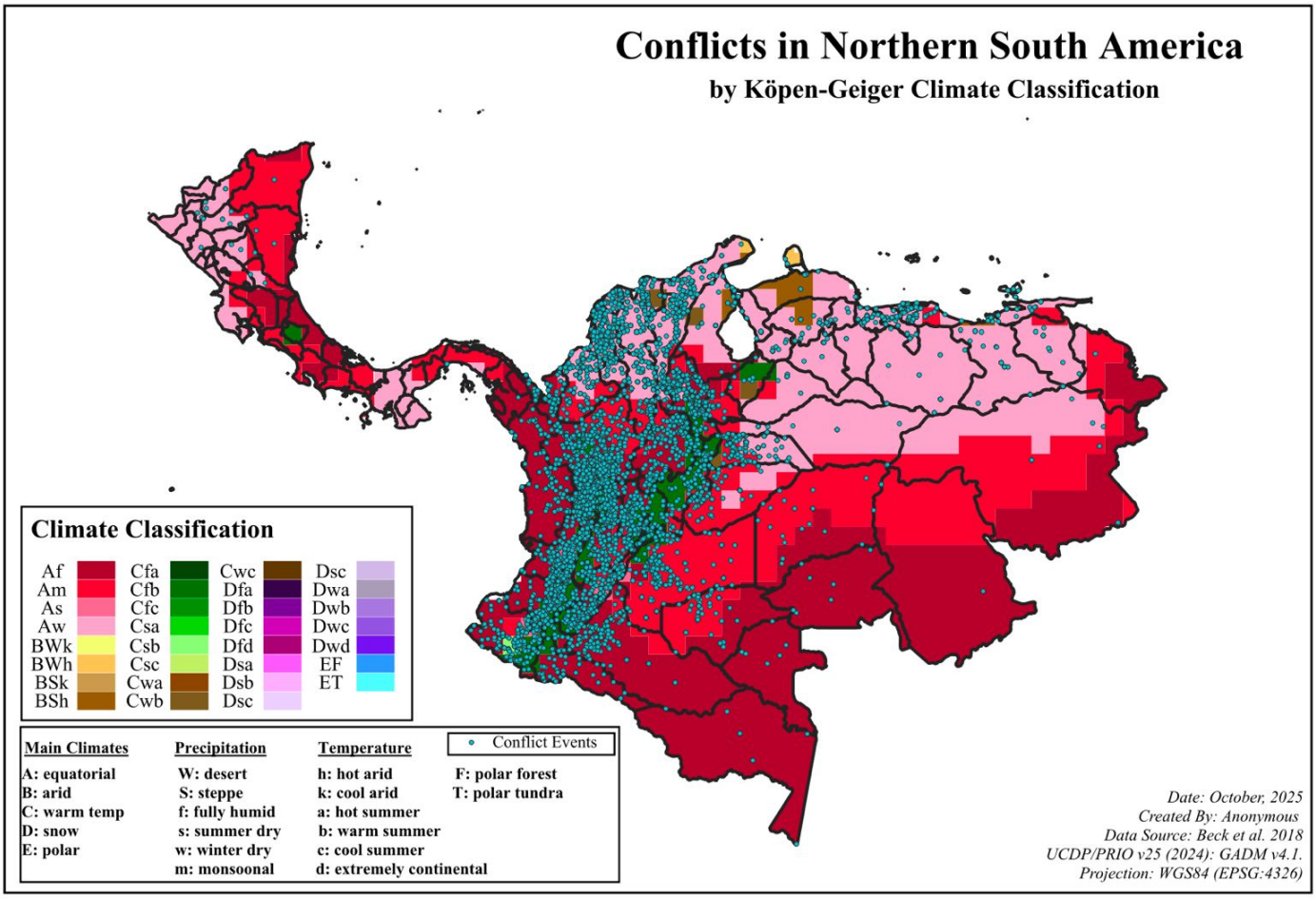
Figure 1